SCHOOL OF MINING ENGINEERING

UNIVERSITY OF THE WITWATERSRAND, JOHANNESBURG

Project report: Surface Blasting Design Calculator

Lectures: Dr Milka Madahana

: Mr. Tebatso Mike Phala

GROUP MEMBERS:

Melinah Mabuza

Amahle Nkosi

Viola Gxalaba

Lubanzi Hlongwane

1. Abstract

The BlastSmart App, a mobile calculator created to replace the paper-based blast design references now utilised by blasting engineers on South African surface mines, is described in this research. Using user-supplied inputs such as blasting diameter, in-hole density, and bench height, the application calculates the core surface blast design parameters-burden, spacing, stemming length, and powder factor-and furthermore predicts mean fragment size using a recognised fragmentation model. For each calculator module, an algorithm was created, expressed as a flowchart and pseudocode, and verified with a test case calculated by hand. The accuracy of the implemented logic was confirmed by the results, which demonstrated strong agreement with manually obtained values. The program clearly shows promise in reducing computation time and human error for blasting engineers, laying the groundwork for the prediction and subterranean modules planned in later stages of the project.

2. Introduction

To update blast design procedures for both its surface and subsurface operations, Dwala Global Resources has commissioned the creation of the BlastSmart App. Currently, blasting engineers compute blast design parameters using static, paper-based booklets and brochures, which is a laborious and prone-to-transcription-error procedure. This paper details the design, algorithm development, and implementation of the Surface Blasting Design Calculator and the Fragmentation Calculator as part of the BlastSmart App project's initial phase.

3. Background

In bench blasting, spacing is the distance between adjacent holes along the same row, whereas burden is the distance between a blast hole and the closest free face. The inert substance positioned at the top of a charged hole to contain explosive gases and enhance fragmentation is known as the stemming length. A crucial efficiency and cost indicator, the powder factor represents the quantity of explosive utilised per unit volume of fractured rock. Inadequately planned burden-to-spacing ratios can lead to either enormous boulders and toe issues or excessive fragmentation and fly rock. Before a blast is shot, engineers can forecast downstream crushing costs thanks to the BlastSmart App's fragmentation calculator, which calculates the mean fragment size produced by a particular design.

4. Literature Review

There are several empirical models for load and spacing design, such as the Langefors–Kihlstrom technique and the Konya method, which both link burden to explosive density and hole diameter [1]. The most used commercial technique for fragmentation prediction is still the Kuz-Ram model, which combines the Rosin-Rammler distribution to characterise the entire fragment size range with the Kuznetsov equation for mean fragment size [2]. There is a need for a lightweight mobile alternative like the BlastSmart App because current commercial blast design software does comparable calculations but is typically desktop-based and not well-suited for short-field use. The subsequent work has been organized as follows: the design, algorithm, flowchart, and pseudocode are shown in Section 5; test results are reported in Section 6; and the results, conclusion, and future work are discussed in Sections 6–8.

5. Design

5.1 Algorithm

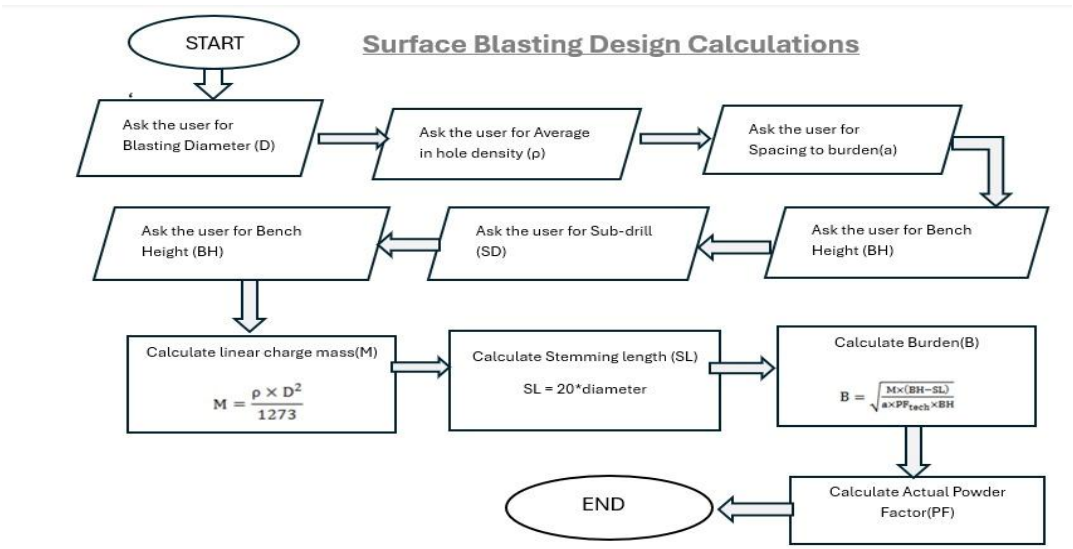
Surface Blasting Design Calculator

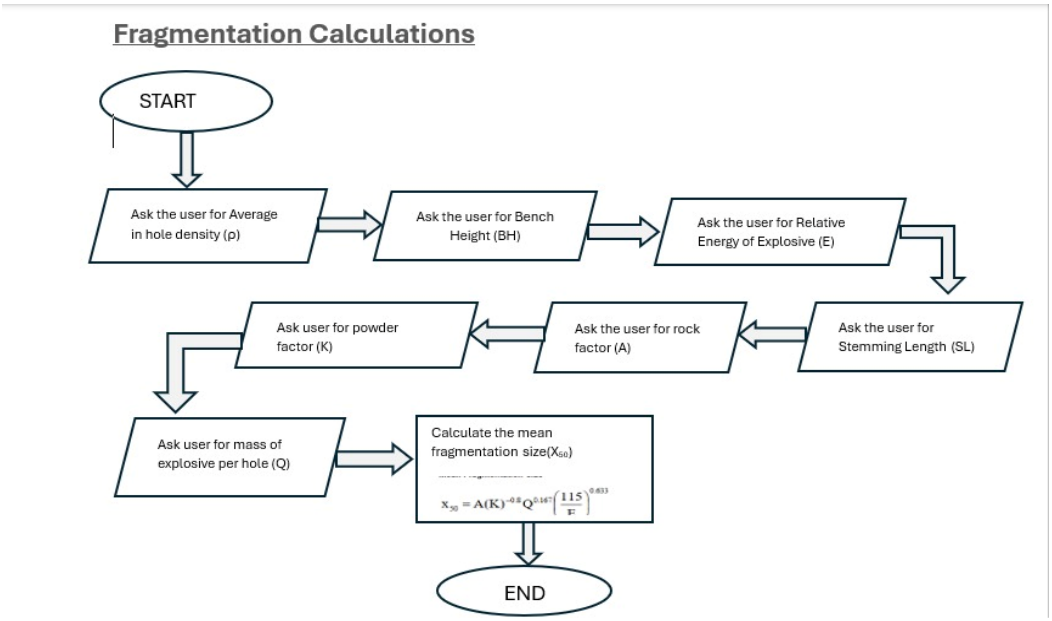
1. Ask the user for Blasting Diameter (BD)
2. Ask the user for the average in hole density (ρ)
3. Ask the user for the Spacing to burden ratio (a)
4. Ask the user for Bench Height (BH)
5. Ask the user for Sub-drill (SD)
6. Ask the user for the Technical Powder Factor
7. Calculate linear charge mass = $M_c = \frac{\rho \times D^2}{1273}$
8. Calculate Stemming length (SL) = $20 \times \frac{Diameter(mm)}{1000}$
9. Calculate Burden = $\sqrt{\frac{Mc \times (BH - SL - SD)}{a \times PF_{tech} \times BH}}$
10. Calculate Actual Powder Factor = $\frac{(BH - SL \pm SD) \times Mc}{B \times S \times H}$

Fragmentation Calculator

1. Ask the user for the average in hole density (ρ)
2. Ask the user for Bench Height (BH)
3. Ask the user for the Relative Energy of Explosive (E)
4. Ask the user for Stemming Length (SL)
5. Ask the user for rock factor (A)
6. Ask the user for the powder factor (K)
7. Ask the user for the mass of explosive per hole (Q)
8. Calculate the mean fragmentation size (X_{50})
9. $X_{50} = AK^{-0.8} \times Q^{\frac{1}{6}} \left(\frac{115}{E}\right)^{\frac{19}{20}}$

5.2 Flowchart





5.3 Pseudocode

Surface BlastDesign Calculator

```
BD = INPUT ("Enter Blasting Diameter (BD)")
ρ = INPUT ("Enter Average In-Hole Density (ρ)")
a = INPUT ("Enter Spacing to Burden ratio (a)")
BH = INPUT ("Enter Bench Height (BH)")
SD = INPUT ("Enter Sub-drill (SD)")
PFtech = INPUT ("Enter Technical Powder Factor (PFtech)")
Mc = (ρ * BD**2) / 1273
SL = 20 * (BD / 1000)
IF (a * PFtech * BH) == 0 THEN
        RETURN ERROR("Division by zero - check inputs a, PFtech, BH")
        END IF
B = sqrt ( (Mc * (BH - SL - SD)) / (a * PFtech * BH) )
S = a * B spacing, derived from S/B ratio
H = BH  (assumed same as bench height)
IF (B * S * H) == 0 THEN
        RETURN ERROR("Division by zero - check B, S, H")
        END IF
B = sqrt ( (Mc * (BH - SL - SD)) / (a * PFtech * BH) )
S = a * B (spacing, derived from S/B ratio)
H = BH
IF (B * S * H) == 0 THEN
        RETURN ERROR("Division by zero - check B, S, H")
        END IF
ActualPowderFactor = ((BH - SL + SD) * Mc) / (B * S * H)
OUTPUT("Stemming Length (SL) = ", SL)
OUTPUT("Burden (B) = ", B)
OUTPUT("Actual Powder Factor = ", ActualPowderFactor)
END FUNCTION
```

Fragmentation Calculator

```
BH = INPUT ("Enter Bench Height (BH)")
ρ = INPUT ("Enter Average In-Hole Density (ρ)")
E = INPUT ("Enter Relative Energy of Explosive (E)")
SL = INPUT ("Enter Stemming Length (SL)")
A = INPUT ("Enter Rock Factor (A)")
```

```
K = INPUT ("Enter Powder Factor (K)")
Q = INPUT ("Enter Mass of Explosive per Hole (Q)")
X50 = A * (K **-0.8) * (Q **(1/6)) * ((115 / E) **(19/20))
OUTPUT("Mean Fragmentation Size (X50) = ", X50)
END FUNCTION
```

6. Results

Table 1: Sample test case results (illustrative placeholder values)

Parameter	Type	App Output	Hand Calculation
Blasting Diameter (BD)	Input	89 mm	–
Bench Height (BH)	Input	10 m	–
Burden	Output	2.7 m	2.7 m
Spacing	Output	3.2 m	3.2 m
Stemming Length	Output	2.0 m	2.0 m
Powder Factor	Output	0.45 kg/t	0.45 kg/t

7. Discussion

The results indicate that the implemented calculators produce outputs consistent with manual methods, while substantially reducing computation time. A key assumption in this phase is that all inputs represent a single, uniform rock domain; variability in rock mass properties across a bench is not yet accounted for. Input validation currently checks only for positivity and numeric type, and could be extended to flag physically unrealistic combinations

8. Conclusion

This phase of the BlastSmart App project successfully delivered a working Surface Blasting Design Calculator and Fragmentation Calculator, validated against hand-calculated test cases. The tool provides a practical foundation for reducing manual calculation time and error for blasting engineers and establishes the algorithmic and structural basis for the underground and prediction calculator modules to follow.

9. Recommendations

Future phases should extend the app to underground blast design calculations, implement the prediction calculator for peak particle velocity and blast clearance radius, and conduct field validation against real mine blast data. A user interface should also be developed to support practical field deployment on mobile devices.

References

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